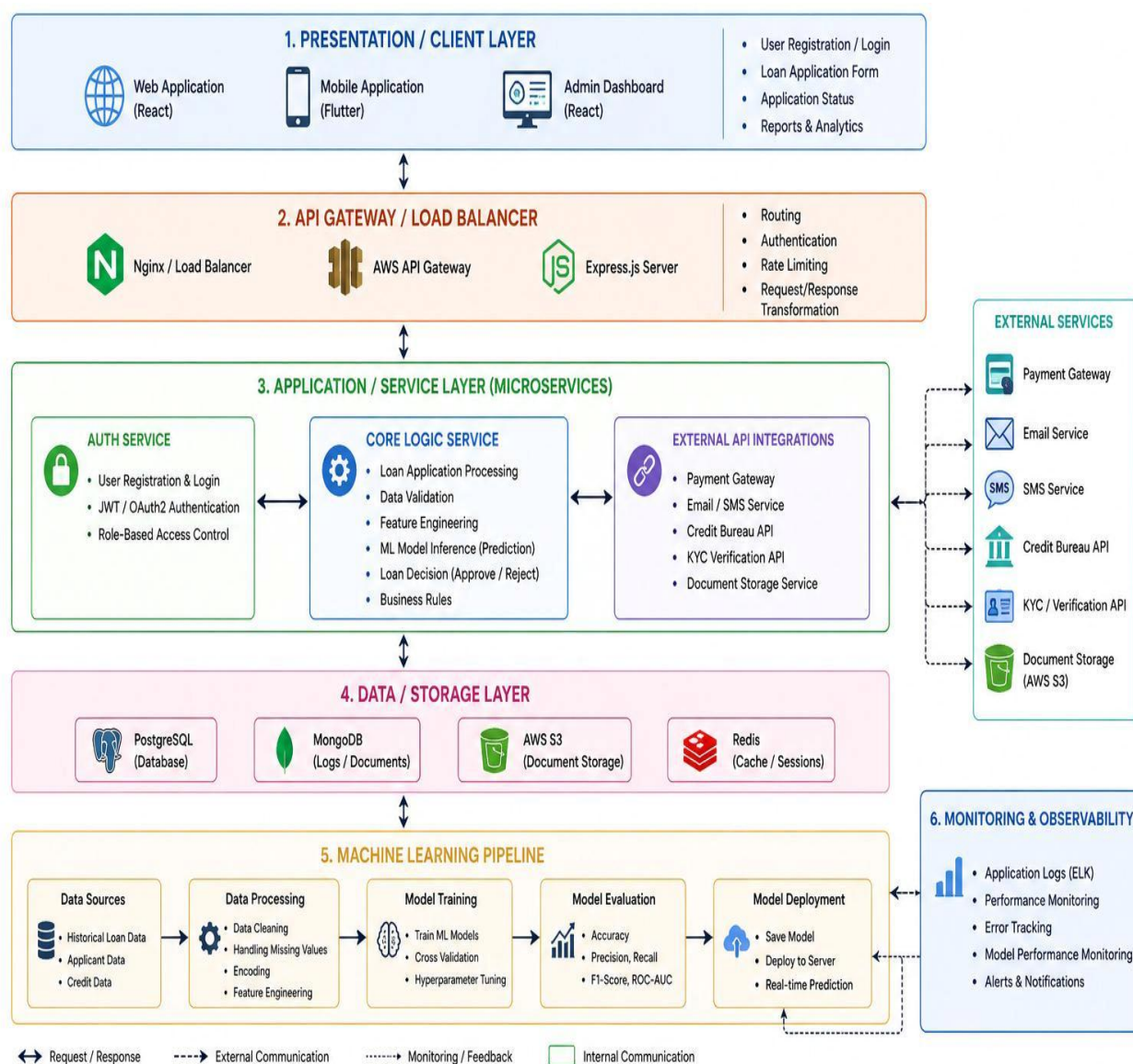


Solution Architecture

Date	27 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5Marks

Solution Architecture Diagram:



Component Description Table:

Component Name	Description / Role in Architecture	Technologies Used
 Presentation Layer	Provides the user interface where customers submit loan applications, view loan status, and administrators monitor applications and prediction results.	• React (Web App) • Flutter (Mobile App) • HTML, CSS, JavaScript • Bootstrap / Material UI
 API Gateway / Load Balancer	Receives client requests, routes them to appropriate services, handles authentication, rate limiting, and request/response transformation.	• Nginx (Load Balancer) • AWS API Gateway / Express Server • HTTPS, RESTful API • JSON
 Microservices / Core Logic Service	Processes loan applications, validates data, performs feature engineering, uses the trained machine learning model to predict loan approval, and applies business rules to generate final decision.	• Python, Flask • Scikit-learn, Pandas, NumPy • Joblib / Pickle (Model Loading) • RESTful Microservices
 Database / Data Storage Layer	Stores user data, loan applications, prediction results, logs, and model files for future reference and reporting.	• PostgreSQL / MySQL (Structured Data) • MongoDB (Logs / Documents) • AWS S3 (Document & Model Storage) • Redis (Cache / Sessions)
 Machine Learning Module	Handles end-to-end machine learning pipeline including data collection, preprocessing, model training, evaluation, and deployment.	• Scikit-learn, XGBoost, Pandas, NumPy • Jupyter Notebook • MLflow / Joblib (Model Management) • Python
 External Services Integrations	Integrates with third-party services for payments, communication, verification, and credit checks.	• Payment Gateway (Razorpay / Stripe) • Email Service (SMTP / SendGrid) • SMS Service (Twilio) • Credit Bureau / KYC Verification APIs